

Cluster-by-Cluster (CbC) Sampling in RAMSES-yOMP

Notes on implementation, feedback coupling, and the missing M^{-2} tail

Branch: `clusterBclusters`

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1 Aim of the project

Standard RAMSES star formation converts gas into “star particles” of a single fixed mass $m_\star$, drawn stochastically (Poisson) from the local Schmidt-type star formation rate. Each particle is not a single star: it represents an unresolved sub-population. Feedback (SN number, energy, mass and metal return) is then injected using a single, global efficiency `eta_sn`, applied identically to every particle regardless of its mass. This implicitly assumes every particle fully samples the stellar IMF, which is a poor approximation once $m_\star$ approaches or falls below the mass needed to sample the high-mass (SN-progenitor) end of the IMF — a regime that is common at high redshift and/or coarse mass resolution.

The Cluster-by-Cluster (CbC) branch replaces this with a physically motivated alternative: each star formation (SF) event is represented not as n identical low-mass particles, but as a small population of *stellar clusters*, with

- cluster masses drawn from a distribution consistent with turbulent-fragmentation / cluster-formation physics, and
- a supernova efficiency `eta_sn` computed *per cluster*, self-consistently from explicit IMF integration, rather than assumed constant.

The physical payoff is that low-mass clusters — which under-sample the massive/SN end of the IMF — are given a correspondingly lower and more stochastic SN efficiency, while massive clusters approach the fully-sampled IMF value. This couples the mass at which star formation is resolved to the strength and timing of the feedback it produces.

2 What is implemented, and how clusters are made

2.1 The star formation budget is unchanged

The local multi-free-fall Krumholz–McKee-type star formation rate (`sf_model=1`) is untouched. Over a hydro sub-step, a cell’s expected star-forming mass is Poisson-sampled into n quanta of $m_\star$:

$$M_{\text{sf}} = n \cdot m_\star \cdot \text{scale_msun}$$

(`star_formation.f90:314, :1236--1238`). M_{sf} is the fixed mass budget for the SF event — it is *not* altered by CbC; CbC only changes what that budget is converted *into*.

2.2 Partitioning M_{sf} into clusters

Two interchangeable kernels (`sf_cluster_kernel`) decompose M_{sf} into one or more cluster masses that sum exactly to M_{sf} :

CMF kernel (`sample_cmf_clusters`, `imf_commons.f90:197--246`). Draws from a power-law cluster mass function $dN/dM \sim M^{-2}$ via inverse-CDF sampling,

$$m_{\text{cl}} = \left[\frac{1}{m_{\text{min}}} - u \left(\frac{1}{m_{\text{min}}} - \frac{1}{m_{\text{max}}} \right) \right]^{-1}, \quad u \sim \mathcal{U}(0, 1),$$

between environment-dependent caps $m_{\text{min,eff}}$, $m_{\text{max,eff}}$ tied to the cell's Jeans mass M_J . Draws are made sequentially, subtracting each cluster mass from a running remainder M_{rem} , until $M_{\text{rem}} < m_{\text{min,eff}}$; the leftover is folded into the last cluster to conserve mass exactly.

Lognormal kernel (`sample_lognormal_clusters`, `imf_commons.f90:262--313`). Same sequential/remainder algorithm, but draws

$$\ln m_{\text{cl}} \sim \mathcal{N}(\ln M_{\text{cl,char}}, \sigma_{\ln M}^2).$$

$M_{\text{cl,char}}$ and $\sigma_{\ln M}$ are computed once per SF event (`star_formation.f90:1100--1113`), reusing the same turbulent density-PDF quantities (s_{igs} , s_{crit}) already computed for the KM star formation rate itself:

$$\rho_{\text{frag}} = d e^{1.5 s_{\text{igs}}} \frac{\text{erfc}\left(\frac{s_{\text{crit}} - 2s_{\text{igs}}}{\sqrt{2s_{\text{igs}}}}\right)}{\text{erfc}\left(\frac{s_{\text{crit}} - s_{\text{igs}}}{\sqrt{2s_{\text{igs}}}}\right)} \quad (\text{SF-weighted core density}) \quad (1)$$

$$R_{\text{char}} = \left(\frac{\pi G \rho_{\text{frag}} dx^{2p}}{5 \sigma^2} \right)^{1/(2p-2)} \quad (\text{virial-balance coherence scale, } \alpha_{\text{vir}}(R_{\text{char}}) = 1) \quad (2)$$

$$M_{\text{g,char}} = \frac{4}{3} \pi \rho_{\text{frag}} R_{\text{char}}^3 \quad (\text{core gas mass}) \quad (3)$$

$$M_{\text{cl,char}} = \text{sf_cluster_eps_cl} \times M_{\text{g,char}} \quad (\text{core} \rightarrow \text{star efficiency, fixed constant, currently 0.3}) \quad (4)$$

$$\sigma_{\ln M} = \sqrt{s_{\text{igs}}} \quad (\text{width} = \text{turbulent density-PDF variance}) \quad (5)$$

Here $p = \text{sf_cluster_turb_index}$ is the Larson index in $\sigma(R)^2 \sim R^{2p}$. Physically, $M_{\text{g,char}}$ is the mass contained in a marginally self-gravitating turbulent core at the SF-weighted density: it is the *local, intrinsic* fragment scale that the cell's own turbulence and self-gravity would produce, independent of how much gas this particular episode happens to convert.

2.3 The IMF/SN-efficiency table

At startup, `init_imf_table` (`imf_commons.f90:128--162`) tabulates $\eta_{\text{SN}}(M_{\text{cl}})$ and $\langle M_{\text{SNII}} \rangle(M_{\text{cl}})$ over `[sf_cluster_mmin, sf_cluster_mmax]` by:

1. integrating the Chabrier (2003) IMF, lognormal below $1 M_{\odot}$ and a power law ($\alpha = 2.35$) above;

2. applying the Weidner–Kroupa (2004) maximum-stellar-mass–cluster-mass relation $m_{\max}(M_{\text{cl}})$, solved by bisection from the condition that one star more massive than $m_{\max}$ is expected per cluster of mass M_{cl} ;
3. integrating star and SN-progenitor counts/masses up to $m_{\max}(M_{\text{cl}})$.

`eta_sn_cluster(mcl, eta, msnii)` then linearly interpolates this table in $\log M_{\text{cl}}$ at feedback time. No `eta_sn` value is stored on the particle; it is recomputed on the fly from the particle’s own mass whenever feedback fires.

3 What the cluster mass implies for feedback

In the legacy path, every particle uses the same constant `eta_sn` (0.313 in this run’s namelist) $\times$ particle mass to set the number of SNe, independent of particle mass.

In the CbC path, `eta_sn_cluster` is called wherever feedback is injected:

- `pm/mechanical_fine.f90:265: snII_freq_cbc = eta_sn_cluster(mass0)` replaces the global `snII_freq` used to set SN timing/number in the mechanical feedback scheme (`mechanical_feedback=.true.` in this run).
- `pm/feedback.f90:399--401`: same substitution for the thermal/kinetic feedback path.

Because $\eta_{\text{SN}}(M_{\text{cl}})$ comes directly from the Weidner-Kroupa $m_{\max}(M_{\text{cl}})$ relation, low-mass clusters (near `sf_cluster_mmin` = $10^3 M_{\odot}$) have a lower $m_{\max}$, sample less of the SN-progenitor tail of the IMF, and get a correspondingly lower η_{SN} ; massive clusters (near `sf_cluster_mmax` = $10^6 M_{\odot}$) approach the fully-sampled IMF value. SN energy, mass and metal-yield injection therefore vary cluster-by-cluster, reflecting each cluster’s own stochastically sampled IMF rather than an assumed universal, fully-sampled population. This mass-efficiency coupling is the physical purpose of the whole exercise — which is precisely why the mass distribution the clusters are actually drawn from matters.

4 Why the resulting mass function is not an M^{-2} tail

The measured cluster mass histogram (log-spaced bins, 10^3 – $10^7 M_{\odot}$) shows a roughly flat pedestal from 10^3 up to $\sim 3 \times 10^4 M_{\odot}$, a sharp spike of ~ 300 counts in the bin centered near $3 \times 10^4 M_{\odot}$, and essentially nothing above that. This is not a lognormal, and it is not a declining power law. The mechanism traced through the code is as follows.

4.1 $M_{\text{cl, char}}$ is decoupled from M_{sf}

$M_{\text{cl, char}}$ (Eq. ??) is computed purely from cell-local quantities (d, σ^2, c_s^2, dx) — it does not reference the event’s own budget M_{sf} at all. M_{sf} , on the other hand, is dominated by the smallest Poisson quantum ($n = 1$, i.e. a single $m_{\star}$) for the large majority of SF events, simply because most cells cross the SF threshold with only marginally enough gas to convert one quantum per sub-step. Since many cells share similar local turbulent conditions at a given AMR level, $M_{\text{cl, char}}$ is also nearly the same across most events.

4.2 The sequential sampler collapses to a single cluster

Both kernels draw sequentially and clip each draw to the remaining budget:

```
mcl = exp(lnMcl_char + sigma_lnM*GaussNum)
mcl = min(max(mcl, mmin), mmax)
mcl = min(mcl, M_rem) ! imf_commons.f90:297
```

Whenever the first draw is comparable to or larger than $M_{\text{rem}} = M_{\text{sf}}$ (expected whenever $M_{\text{cl, char}} \gtrsim M_{\text{sf}}$, per §4.1), the clip collapses the entire episode to *one* cluster with mass exactly M_{sf} — discarding the kernel’s shape for that event entirely. Because this happens for the dominant, nearly-identical $n = 1$ events, the outcome is a narrow spike at (close to) the single-quantum mass $m_\star \cdot \text{scale_msun}$: this is the observed spike, and it is a signature of the stochastic SF quantization leaking through, not of cluster-formation physics.

4.3 The minority of fragmenting events produce a flat, not power-law, tail

The less common events with larger M_{sf} (larger n) do fragment into a handful of clusters. But two further effects prevent this sub-population from converging to a smooth M^{-2} -like envelope:

1. **No enforced fragment count.** Because $M_{\text{cl, char}}$ is not tied to M_{sf} , the expected number of fragments $N_{\text{frag}} \sim M_{\text{sf}}/M_{\text{cl, char}}$ is never an explicit, designed quantity — it is an accidental side effect of how many times the clip in §4.2 is avoided before M_{rem} runs out. Aggregating many events therefore does not correspond to summing many properly-normalized per-event CMFs/lognormals over a meaningful spread of $M_{\text{cl, char}}/M_{\text{sf}}$ ratios.
2. **Width mismatch.** $\sigma_{\ln M} = \sqrt{s_{\text{igs}}}$ (Eq. ??) inherits the full variance of the turbulent *density* PDF directly. Gravo-turbulent fragmentation theory (e.g. Padoan & Nordlund 2002; Hennebelle & Chabrier 2008) predicts a substantially *narrower* core mass function width after the density-to-mass mapping ($M \sim \rho R^3$, with R itself density-dependent through Eq. ??) is applied. Using the raw density-PDF width over-disperses individual draws, washing out any convergent shape when many events are combined — consistent with the observed flat pedestal rather than a peaked or power-law distribution.

4.4 Summary of the diagnosis

The spike is the stochastic SF particle-mass quantization ($n = 1$ events) showing through unfragmented; the flat pedestal is the minority of multi-cluster events, under-dispersed in count and over-dispersed in width relative to what the physics intends. Neither reflects the intended turbulence-driven cluster mass function.

5 Proposed fix (discussed, not yet implemented)

Tie $M_{\text{cl, char}}$ explicitly to the event’s own budget rather than treating it as an independent absolute scale:

$$\epsilon \equiv \min\left(\frac{M_{\text{frag}}}{M_{\text{sf}}}, 1\right), \quad M_{\text{cl, char}} = \epsilon M_{\text{sf}} = \min(M_{\text{frag}}, M_{\text{sf}}),$$

where $M_{\text{frag}} \equiv M_{\text{g, char}}$ (Eq. ??) is kept as the existing, physically-derived local fragment scale (turbulence- and, via c_s^2 , metallicity/cooling-aware). This guarantees:

- $M_{\text{cl, char}} \leq M_{\text{sf}}$ by construction — the collapse to a single cluster becomes a genuine physical regime ($M_{\text{sf}} \lesssim M_{\text{frag}}$, i.e. the budget does not even cover one local fragment), not an artifact of comparing two unrelated scales;
- $N_{\text{frag}} \sim 1/\epsilon = M_{\text{sf}}/M_{\text{frag}}$ becomes an explicit, physically meaningful quantity that varies smoothly with each event’s own budget, instead of being dominated by the Poisson SF quantization;
- the fixed constant `sf_cluster_eps_cl`= 0.3 is replaced by a per-event, physically-derived ratio.

Separately, $\sigma_{\ln M}$ should be revisited: propagate s_{igs} through the density→mass mapping (Eqs. ??–??) rather than adopting $\sqrt{s_{\text{igs}}}$ directly, so the sampled width reflects the core mass function width predicted by fragmentation theory rather than the raw turbulent density-PDF width.